

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	SWTID-2026-4457
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Script-controlled ACL for restricting record access based on field value (ServiceNow)

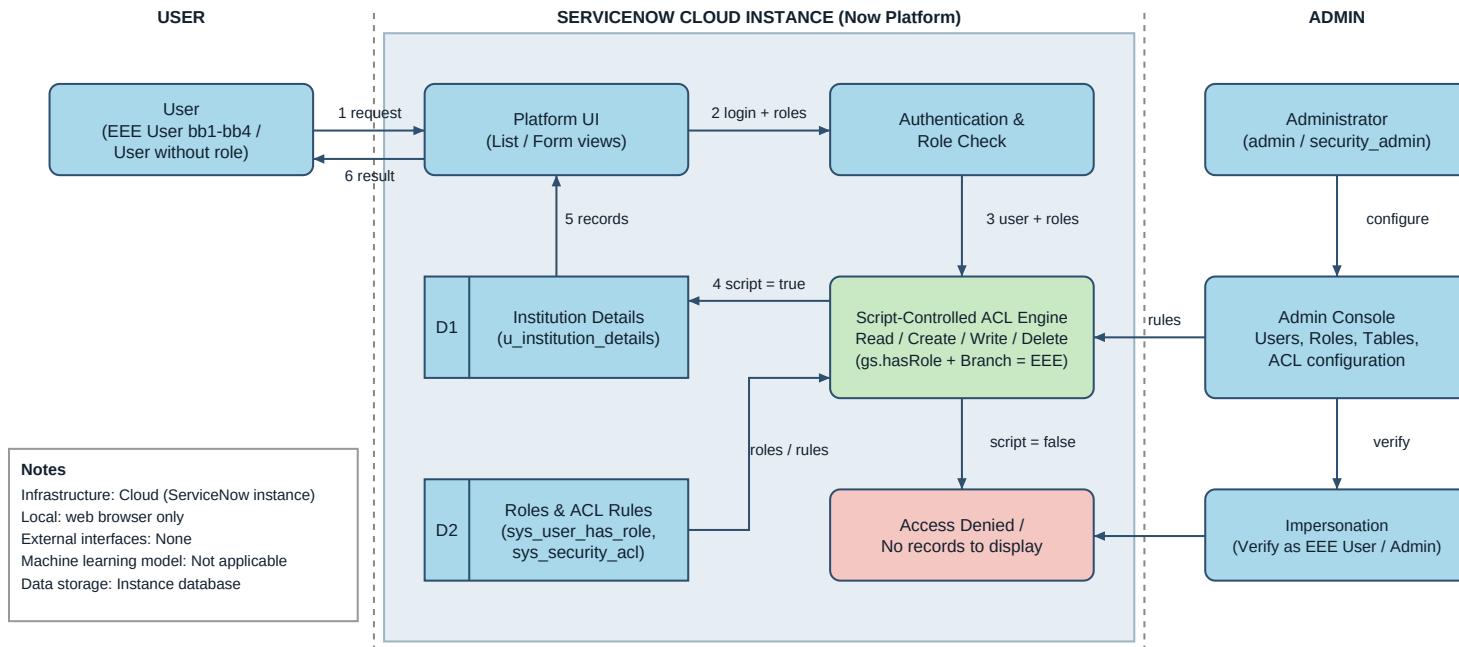


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Users (EEE User, Admin) interact with the Institution Details list and form through the ServiceNow web interface; the New button and edit options appear only for permitted roles.	ServiceNow Platform UI (browser: HTML, CSS, JavaScript)
2.	Application Logic-1	Authentication and role check: user logs in and the roles assigned (bb1, bb2, bb3, bb4, admin) are read.	ServiceNow User Administration, Roles, gs.hasRole()
3.	Application Logic-2	Script-controlled Read ACL (Advanced): data condition Branch is EEE; script returns true for admin or bb1, otherwise false.	ServiceNow ACL, JavaScript (ECMAScript 2021 mode)
4.	Application Logic-3	Create, Write and Delete ACLs that require roles bb2, bb3 and bb4 respectively.	ServiceNow ACL (Allow If, Requires role)
5.	Database	Custom table u_institution_details with Student Roll Number, Student Name, Faculty Name, Branch (ECE, EEE, CSE), Email, Phone Number, Description.	ServiceNow platform database (custom table)
6.	Cloud Database	Records, users, roles and ACL rules are stored in the cloud-hosted ServiceNow instance.	ServiceNow cloud instance database
7.	File Storage	No separate file storage is required; all data is held in the instance.	ServiceNow instance storage
8.	External API-1	Not used in this project.	N/A
9.	External API-2	Not used in this project.	N/A
10.	Machine Learning Model	Not applicable to this project.	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on cloud. Local Server Configuration: web browser only. Cloud Server Configuration: ServiceNow-hosted instance (Global application scope).	ServiceNow Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The platform is ServiceNow; ACL logic is written in standard JavaScript. No separate open-source framework is used.	JavaScript (ES12), ServiceNow Glide API
2.	Security Implementations	Record-level Read, Create, Write and Delete ACLs; role-based access control; data condition (Branch is EEE); default deny (return false); security_admin elevation required to edit ACLs.	ServiceNow ACLs, RBAC, gs.hasRole()
3.	Scalable Architecture	Cloud multi-tier platform; more roles, branches, users and records can be added by creating roles and ACLs or updating the script and data condition.	ServiceNow Now Platform
4.	Availability	The solution is available whenever the cloud-hosted ServiceNow instance is available; no extra servers are managed by the project team.	ServiceNow cloud infrastructure
5.	Performance	ACLs use simple role checks and a field condition evaluated on the server, so list views load quickly for the small Institution Details table.	Server-side ACL evaluation (ServiceNow)

References:

<https://c4model.com/>

<https://docs.servicenow.com> (Access Control Rules, Script-controlled ACLs)